1.2 Classwork: Arithmetic Sequences — Markscheme

1. [Maximum mark: 8]

   The sequence $7,\ 11,\ 15,\ 19,\ \ldots$

   (a) $u_1 = 7$ **A1**

   (b) The sequence is arithmetic. **A1**

   Reason: the difference between consecutive terms is constant: $11 - 7 = 15 - 11 = 19 - 15 = 4$. **R1**

   Accept "yes", "arithmetic", or equivalent for the A1.

   R1 is awarded for any statement that the difference between consecutive terms is the same (in words, or by showing at least two equal differences). "It goes up by 4 every time" is sufficient. Award the R1 if the reasoning is shown in a later part: a candidate who states only that the sequence is arithmetic here, and then computes the equal differences under (c), has given the reason and earns the R1.

   (c) $d = 4$ **A1**

   (d) $u_5 = 23$, $u_6 = 27$ **A1**

   Both values required for the single mark. Accept "$23, 27$*" written as a continuation of the list.*

   FT: award A1 for the next two terms obtained by adding the candidate's own d *from (c) to* 19.

   (e) $u_n = u_1 + (n-1)d = 7 + (n-1)(4)$ **M1**

   $u_n = 4n + 3$ **A1**

   Accept the unsimplified form $7 + 4(n-1)$ *for full marks.*

   FT: award M1A1 for a correct expression formed from the candidate's own u_1 *and* d.

   (f) $u_{50} = 4(50) + 3 = 203$ **A1**

   FT: award A1 for the value obtained by substituting $n = 50$ *into the candidate's own expression from (e).*

2. [Maximum mark: 6]

$u_1 = 40$, $d = -3$.

(a) $40,\ 37,\ 34,\ 31$ **A1**

All four terms required for the mark.

(b) $u_{20} = 40 + (20-1)(-3) = 40 - 57$ **M1**

$u_{20} = -17$ **A1**

M1 is for any correct use of $u_n = u_1 + (n-1)d$ *with* $n = 20$*, however the arithmetic is then arranged; a GDC sequence table showing the* 20^{th} *term also earns the M1.*

(c) $40 + (n-1)(-3) = -50$ **M1**

$(n-1)(-3) = -90$

$n - 1 = 30$ **A1**

$n = 31$ **A1**

Accept the equivalent set-up $43 - 3n = -50$ *leading to* $3n = 93$*.*

The A1 for $n - 1 = 30$ *is for any correct intermediate line that isolates* n *(e.g.* $-3n = -93$*); it does not have to be written in this exact form.*

A candidate who states $n = 31$ *from a GDC table with no algebraic working earns the final A1 only, since the question asks for algebraic working.*